

# Linear Algebra: linear functionals

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June 9, 2026

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## Abstract

This page reviews Linear Algebra, from the perspective of linear functionals. We deem that readers already know the basic facts and main results of linear algebra: unique representation of vectors with respect to a basis, determinant, eigenvectors and eigenspaces, direct sum and projection, inner product and the spectral decomposition.

## 1 Preliminaries

Recall that: the philosophy of basis.

**Lemma 1.1.** Let  $V$  be a Vector Space over the field  $F$ . A subset  $\beta \subset V$  is a basis for  $V$  if and only if:  
For any vector  $x \in V$ , there exist unique finite non-zero scalars  $a_i \in F$  and distinct vectors  $v_i \in \beta$  such that

$$x = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

The 'unique' implies that:

if  $x = a_1 v_1 + \cdots + a_n v_n = b_1 w_1 + \cdots + b_m w_m$  where  $a_i, b_i \in F \setminus \{0\}$  and  $v_i, w_j \in \beta$  are distinct within each sum, then  $n = m$  and for some permutation  $\sigma$ ,  $v_i = w_{\sigma(i)}$ ,  $a_i = b_{\sigma(i)}$  for all  $1 \leq i \leq n$ .

Since the representation is unique, we can define the *coordinate representation* with respect to the basis  $\beta$ .

**Definition 1.2.** Let  $\beta = \{v_1, \dots, v_n\}$  be an ordered basis for  $V$ .  
The *coordinate representation* (or *coordinate map*) with respect to  $\beta$  is defined as:

$$[\cdot]_{\beta} : V \rightarrow F^n : x \mapsto \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

where  $x = \sum_{i=1}^n a_i v_i$ .

## 2 Dual Space

Let  $V$  be a vector space over a field  $F$ .

**Definition 2.1.** A linear map  $f : V \rightarrow F$  is called a *linear functional* on  $V$ .

**Example.** Let  $F[x]$  be the set of polynomials over  $F$ . For each  $\alpha \in F$ , define

$$L_\alpha : F[x] \rightarrow F : p(x) \mapsto p(\alpha)$$

This is called the 'evaluation' at  $\alpha$ . Clearly, it is a linear functional since

$$L_\alpha(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L_\alpha(p(x)) + L_\alpha(q(x))$$

**Example.** Let  $C([a, b])$  be the space of continuous real-valued functions on  $[a, b]$ . Define

$$L : C([a, b]) \rightarrow \mathbb{R} : f \mapsto \int_a^b f(t) dt$$

$L$  is clearly a linear functional by the linearity of the integral.

**Definition 2.2.** The collection of all linear functionals on  $V$ , denoted by  $\mathcal{L}(V, F)$ , is called the *dual space* of  $V$  and is denoted as  $V^*$ .

**Theorem 2.3.** Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\beta = \{\alpha_1, \dots, \alpha_n\}$  be a basis for  $V$ . Then, there exists a unique *dual basis*  $\beta^* = \{f_1, \dots, f_n\}$  for  $V^*$  such that  $f_i(\alpha_j) = \delta_{ij}$ . Furthermore, for each linear functional  $f \in V^*$ ,

$$f = \sum_{i=1}^n f(\alpha_i) f_i$$

and for each vector  $\alpha \in V$ ,

$$\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$$

*Proof.* By the property of basis, for any  $\alpha \in V$ , there exist unique scalars  $a_1, \dots, a_n \in F$  such that

$$\alpha = a_1 \alpha_1 + \dots + a_n \alpha_n$$

For each  $1 \leq i \leq n$ , define  $f_i : V \rightarrow F$  by  $f_i(\alpha) = a_i$  (or  $f_i(\alpha) = ([\alpha]_\beta)_i$ ).

Then  $f_i(\alpha_j) = \delta_{ij}$ . Since  $\dim V = \dim V^*$ , it is enough to show that the set  $\beta^* = \{f_1, \dots, f_n\}$  is linearly independent. Suppose that for some  $c_i \in F$ ,  $\sum_{i=1}^n c_i f_i = 0$ . Then for each  $1 \leq j \leq n$ ,

$$\left( \sum_{i=1}^n c_i f_i \right) (\alpha_j) = \sum_{i=1}^n c_i f_i(\alpha_j) = c_j = 0$$

Thus,  $\beta^*$  is linearly independent and forms a basis for  $V^*$ .

The second assertion can be proved as follows: if  $\alpha \in V$  is represented as  $\alpha = \sum_{i=1}^n x_i \alpha_i$  where  $x_i \in F$ , then for each  $1 \leq j \leq n$ :

$$f_j(\alpha) = f_j \left( \sum_{i=1}^n x_i \alpha_i \right) = \sum_{i=1}^n x_i f_j(\alpha_i) = x_j$$

Substituting  $x_j$  back into the representation of  $\alpha$ , we obtain  $\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$ . □

## Application: Lagrange Interpolation

Let  $P_n(F) = \{p \in F[x] : \deg p \leq n\}$ , and let  $t_0, t_1, \dots, t_n \in F$  be distinct elements. For each  $0 \leq i \leq n$ , define the linear functional  $L_i : P_n(F) \rightarrow F$  by  $L_i(p) = p(t_i)$ . These  $\{L_0, L_1, \dots, L_n\}$  are linearly independent. Suppose that

$$c_0 L_0 + c_1 L_1 + \dots + c_n L_n = 0 \quad \dots (*)$$

where  $c_i \in F$ . By substituting the standard basis  $\{1, x, x^2, \dots, x^n\}$  into  $(*)$ , we obtain a system of equations:

$$\begin{cases} c_0 + c_1 + \dots + c_n = 0 \\ c_0 t_0 + c_1 t_1 + \dots + c_n t_n = 0 \\ \vdots \\ c_0 t_0^n + c_1 t_1^n + \dots + c_n t_n^n = 0 \end{cases} \implies \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_0 & t_1 & \dots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_0^n & t_1^n & \dots & t_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix above is the **Vandermonde Matrix**, which is invertible since  $t_i$  are distinct. Thus  $c_0 = c_1 = \dots = c_n = 0$ . More precisely, define a linear map

$$T : P_n(F) \rightarrow F^{n+1} : p(x) \mapsto (p(t_0), \dots, p(t_n))$$

For  $\beta = \{1, x, \dots, x^n\}$ ,  $[T]_\beta$  is the **Vandermonde Matrix**.

If we choose a different basis  $\beta' = \{1, (x - t_0), (x - t_0)(x - t_1), \dots, \prod_{j=0}^{n-1} (x - t_j)\}$ , then  $[T]_{\beta'}$  becomes:

$$\det[T]_{\beta'} = \det \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & t_1 - t_0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_n - t_0 & \dots & \prod_{j=0}^{n-1} (t_n - t_j) \end{bmatrix} = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0$$

Since the determinant is invariant with respects to the similar matrix, it is proved.

**Theorem 2.4.**  $\{L_0, \dots, L_n\}$  is a dual basis for  $P_n(F)^*$  with respect to the basis  $\{P_0(x), \dots, P_n(x)\}$  for  $P_n(F)$ , where

$$P_i(x) = \prod_{j \neq i} \frac{x - t_j}{t_i - t_j}$$

Since  $L_j(P_i) = P_i(t_j) = \delta_{ij}$ , for any  $p(x) \in P_n(F)$ , we have:

$$p(x) = \sum_{i=0}^n L_i(p) P_i(x) = \sum_{i=0}^n p(t_i) P_i(x)$$

Furthermore, by the linearity of the integral,  $\int_a^b p(x) dx = \sum_{i=0}^n p(t_i) \int_a^b P_i(x) dx$ .

### 3 Annihilators

**Definition 3.1.** Let  $S$  be a subset of a vector space  $V$ . The annihilator of  $S$ , denoted by  $S^0$ , is defined as:

$$S^0 = \{f \in V^* : f(\alpha) = 0 \text{ for all } \alpha \in S\}$$

**Example.** In  $P_2(F)$  with basis  $\beta = \{1, x, x^2\}$  and dual basis  $\{f_0, f_1, f_2\}$  (where  $f_i$  extracts the coefficient of  $x^i$ ), consider  $S = \{x\}$ . Then  $f \in S^0$  if  $f(x) = 0$ . Since  $f = c_0f_0 + c_1f_1 + c_2f_2$ ,  $f(x) = c_1 = 0$ . Thus,  $S^0 = \text{span}\{f_0, f_2\}$ .

**Theorem 3.2.** Let  $V$  be a finite-dimensional vector space and  $W$  be a subspace of  $V$ . Then:

$$\dim W + \dim W^0 = \dim V$$

*Proof.* Let  $\{\alpha_1, \dots, \alpha_k\}$  be a basis for  $W$ . Extend it to a basis  $\{\alpha_1, \dots, \alpha_n\}$  for  $V$ . Let  $\{f_1, \dots, f_n\}$  be the dual basis for  $V^*$ . For  $i \geq k+1$ ,  $f_i(\alpha_j) = 0$  for all  $1 \leq j \leq k$ , so  $f_i \in W^0$ . The set  $\{f_{k+1}, \dots, f_n\}$  is linearly independent. For any  $f \in W^0$ , we can write  $f = \sum_{i=1}^n f(\alpha_i)f_i$ . Since  $f \in W^0$ ,  $f(\alpha_i) = 0$  for  $i \leq k$ . Thus  $f = \sum_{i=k+1}^n f(\alpha_i)f_i$ , which means  $\{f_{k+1}, \dots, f_n\}$  spans  $W^0$ . Therefore,  $\dim W^0 = n - k = \dim V - \dim W$ . □

## 4 Double Dual

*Question: Is every basis for  $V^*$  a dual of some basis for  $V$ ?*

**Lemma 4.1.** Let  $V$  be a finite dimensional vector space over  $F$ . For each  $\alpha \in V$ , define

$$L_\alpha : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, it is linear functional on  $V^*$ . Moreover, if  $L_\alpha(f) = 0$  for any  $f \in V^*$ , then  $\alpha = 0$ .

**Note:** the last assertion can be said:  $(V^*)^0 = \{0\}$ .

*Proof.* Let  $f, g \in V^*$  and  $c \in F$  be given. Then,

$$L_\alpha(cf + g) = (cf + g)(\alpha) = cf(\alpha) + g(\alpha) = cL_\alpha(f) + L_\alpha(g)$$

Thus proved. To show the second assertion, let  $\alpha \neq 0$ . Then, there exists a basis  $\beta$  for  $V$  containing  $\alpha$ . Then, there is a dual basis  $\beta^*$  for  $V^*$ . Take  $f_\alpha \in \beta^*$  corresponding to  $\alpha$ . Then,  $f_\alpha(\alpha) = 1 \neq 0$ . Thus proved.  $\square$

**Theorem 4.2.** Let  $V$  be a finite dimensional vector space over  $F$ . For each  $\alpha \in V$ , define

$$L_\alpha : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, the map  $V \rightarrow V^{**} : \alpha \mapsto L_\alpha$  is an Isomorphism.

*Proof.* Let  $\alpha, \beta \in V$  and  $c \in F$ , and for any  $f \in V^*$ ,

$$L_{c\alpha + \beta}(f) = f(c\alpha + \beta) = cf(\alpha) + f(\beta) = cL_\alpha(f) + L_\beta(f) = (cL_\alpha + L_\beta)(f)$$

Moreover, by the Lemma, the kernel of the map is trivial. Thus injective, and surjective because:

$$\dim V^{**} = \dim V^* = \dim V$$

$\square$

**Corollary 4.3.** Let  $V$  be a finite dimensional vector space over  $F$ .

IF  $L \in V^{**}$ , then there exists a unique  $\alpha \in V$  such that for all  $f \in V^*$ ,  $L(f) = f(\alpha)$ .

**Corollary 4.4.** Let  $V$  be a finite dimensional vector space over  $F$ .

Every basis for  $V^*$  is the dual of some basis for  $V$ .

*Proof.* Let  $\{f_1, \dots, f_n\}$  be a basis for  $V^*$ . Then, there is a dual basis  $\{L_1, \dots, L_n\}$  for  $V^{**}$  such that  $L_i(f_j) = \delta_{ij}$ . By the above corollary, for each  $L_i$ , there exists a unique  $\alpha_i \in V$  such that  $L_i(f) = f(\alpha_i)$  for all  $f \in V^*$ . Now,  $L_i(f_j) = f_j(\alpha_i) = \delta_{ji}$ , the claim is proved.  $\square$

## 5 Transpose

**Theorem 5.1.** Let  $V$  and  $W$  be finite dimensional vector space over  $F$ , and let  $T : V \rightarrow W$  be a linear map. Then,

$$T^t : W^* \rightarrow V^* : g \mapsto g \circ T$$

is a linear map. Furthermore, for any ordered basis  $\beta$  and  $\gamma$  for  $V$  and  $W$ , respectively,

$$([T]_{\beta}^{\gamma})^t = [T^t]_{\gamma^*}^{\beta^*}$$

*Proof.* Since the composition of two linear maps is linear, the map  $T^t$  is well-defined. Given  $f, g \in W^*$  and  $c \in F$ ,

$$T^t(cf + g) = (cf + g) \circ T = cf \circ T + g \circ T = cT^t(f) + T^t(g)$$

Thus, the first assertion is proved. Let  $\beta = \{x_1, \dots, x_n\}$  and  $\gamma = \{y_1, \dots, y_m\}$ . Then, there are dual basis for  $V^*$ ,  $W^*$

$$\beta^* = \{f_1, \dots, f_n\}, \quad \gamma^* = \{g_1, \dots, g_m\}$$

respectively. Observe that: for each  $1 \leq j \leq m$ ,

$$T^t(g_j) = g_j \circ T = \sum_{s=1}^n (g_j \circ T)(x_s) f_s$$

Meanwhile, denote  $A = [T]_{\beta}^{\gamma}$ . Then,

$$(g_j \circ T)(x_i) = g_j(T(x_i)) = g_j\left(\sum_{k=1}^m A_{ki} y_k\right) = \sum_{k=1}^m A_{ki} g_j(y_k) = \sum_{k=1}^m A_{ki} \delta_{jk} = A_{ji}$$

Therefore,

$$[T^t]_{\gamma^*}^{\beta^*} = [[T^t(g_1)]_{\beta^*} \quad \dots \quad [T^t(g_m)]_{\beta^*}] = A^t = ([T]_{\beta}^{\gamma})^t$$

□

**Theorem 5.2.** Let  $F$  be a field, and let  $T : F^n \rightarrow F^m$  be a map. A map  $T$  is linear map if and only if:

$$\text{There exist } T_1, \dots, T_m \in (F^n)^* \text{ such that for all } x \in F^n, T(x) = (T_1(x), \dots, T_m(x))$$

*Proof.* Let  $\{g_1, \dots, g_m\}$  be a dual basis for  $F^m$  with respects to standard basis for  $F^m$ . For each  $1 \leq i \leq m$ , define

$$T_i = T^t(g_i)$$

Let  $x \in V$  be given. Then, for each  $1 \leq i \leq m$ ,

$$T_i(x) = T^t(g_i)(x) = (g_i \circ T)(x) = g_i(T(x)) = ([T(x)]_{\gamma})_i$$

Thus proved. The converse is trivial.

□

## 6 Observation

Let  $V$  and  $W$  be Vector Space over the field  $F$ . If both  $V$  and  $W$  have finite dimension, we can take basis  $\beta$  and  $\gamma$  for  $V$  and  $W$ , respectively. Then, we can identify both  $V$  and  $W$  as cartesian product of the given field as  $F^n$  and  $F^m$ . More precisely, for any basis  $\beta = \{v_1, \dots, v_n\}$ , we can take the Isomorphism as

$$\varphi: V \rightarrow F^n : x \mapsto [x]_\beta$$

that is,  $v_i \mapsto e_i$ , where  $e_i$  is the standard base of  $F^n$ . In this consideration, we assume that  $T: F^n \rightarrow F^m$ . Now,

$$T(x) = (T_1(x), \dots, T_m(x))$$

we can represent as  $T$  to this. In this perspective, arising a natural question, are must the  $T_i$ 's be linear if  $T$  is linear? Conversely, if all  $T_i$ 's are linear, then  $T$  is linear? We have already answered this question, using the language of linear functionals. However, a question still remains: when this  $T$  acts with 'good' behavior?